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Off the List, Not Into the Water: A Competing-Risks Analysis of Learn-to-Swim Waiting Times Across Eleven Council Aquatic Centres, Against an Indicator That Counts Only Placements

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Abstract. A waiting list has two exits, and public reporting on waiting lists conventionally describes only one of them. We analyse 9,486 registrations entering the learn-to-swim waiting lists of eleven council aquatic centres, operated under a single leisure-services contract, over thirty consecutive months. Registrations are followed to placement in a class, to withdrawal, or to administrative censoring at the data cut, and analysed as competing risks. Placement incidence reaches 0.587 at twelve months and effectively plateaus thereafter; withdrawal incidence continues to accrue, reaching 0.341 by twenty-four months. A Kaplan-Meier analysis treating withdrawal as censoring overstates twelve-month placement incidence by 11.3 percentage points. The operator's published median-wait indicator, computed over placed registrations only, fell from 8.9 to 6.2 weeks across the eight quarters studied while twelve-month placement incidence for the same entry cohorts fell from 0.631 to 0.528. At three of eleven centres, withdrawal incidence at twelve months exceeds placement incidence. An ablation removing the sibling-priority term from the subdistribution model moves every remaining coefficient by less than 0.03; we retain it for completeness. Our results suggest that a waiting-time measure conditioned on the outcome it reports may improve as the underlying service deteriorates.

Introduction

A family registers a child for swimming lessons at a council pool and is told they are on a list. Some weeks or months later one of three things is true: they have a place, they have given up, or they are still waiting. Only the first of these is legible to the reporting that the centre, the operator, and the council do about the list.

This asymmetry is not peculiar to aquatic centres. Waiting-time reporting in public services is dominated by measures computed over completed waits [1], [2], and the incentive properties of such measures are well documented [3], [4], [5]. The statistical machinery for the general case is equally well established: where more than one exit is possible, the standard survival estimator [6] is biased for the probability of any one of them, and competing-risks methods should be used instead [7], [8], [9]. The two literatures are rarely brought into contact. A waiting list is a canonical competing-risks setting and is almost never reported as one.

We take a single leisure-services contract covering eleven council aquatic centres and analyse its learn-to-swim waiting lists as time-to-event data with two absorbing states. Our contributions are three, each modest:

- We estimate cumulative incidence functions for placement and withdrawal across a complete registration population, and quantify how far the conventional censoring treatment departs from them.

- We compare the operator's published quarterly median-wait indicator against the competing-risks median for the same entry cohorts, and characterise the divergence over eight quarters.
- We report a subdistribution model of placement whose covariates are drawn from fields the operator's booking system already holds, and an ablation suggesting the model's documented priority term is not doing much work.

We make no claim that any centre in the portfolio is managed badly, and the analysis identifies no decision that could have been taken differently on the information available at the time.

Background

Waiting lists in public services have been studied principally as a management problem: how long people wait, what shortens the wait, and what a published target does to the distribution of waits [1], [5], [10]. The recurring finding is that a measure attached to consequences reshapes the activity it measures rather than the underlying capacity [3], [4]. Our concern is narrower and prior to that literature: what the measure is arithmetically capable of seeing before any behavioural response is considered.

The competing-risks framework supplies the vocabulary. Where a subject may leave a state by more than one route, the cause-specific hazard and the cumulative incidence function answer different questions, and both should be reported [11]. The subdistribution hazard model [12] provides covariate effects on the cumulative incidence scale, with reporting conventions since consolidated [13]; tests for equality of cumulative incidence across groups follow [14]. Proportionality is assessed on the scaled residuals of the corresponding Cox model [15], [16].

Abandonment from queues is treated seriously in one adjacent literature, that of telephone service systems, where the impatient caller is a first-class object of study rather than a data-quality nuisance [17], [18]. Dropout from organised youth sport has likewise been characterised in its own right [19]. That swimming instruction is worth queueing for at all rests on a separate evidence base concerning childhood drowning [20], [21], which we do not attempt to extend.

Prior work at this institution has examined how allocation lists actually clear. An interview study of a community-garden waitlist found that neither self-reported reason nor measured wait time separated households above and below the list's published priority score [22], and an audit of a campus parking-permit overhaul found the replacement priority-tier scheme producing rankings closely correlated with the blanket waitlist it replaced, over a substantially overlapping recipient set [23]. Related work has shown how an indicator denominated in one unit books only the costs that unit can carry [24]. The present study differs in taking the list's non-completions as the object rather than as attrition.

Method

Setting and data

Eleven council aquatic centres across three local government areas are operated under a single leisure-services contract. Each centre maintains one learn-to-swim waiting list, entered by online registration or at the counter, and clears it by offering places as classes turn over. We obtained the operator's registration extract covering thirty consecutive months: 9,486 registrations, each carrying an entry timestamp, a centre, the fields the booking form collects, and an end state.

Registrations end in one of three ways. A **placement** is an accepted offer of a class place. A **withdrawal** is a family-initiated removal from the list, or an operator-initiated removal following two unanswered offers, which the system records identically. Registrations neither placed nor withdrawn at the data cut are administratively censored. Of the 9,486 registrations, 5,102 were placed, 3,187 withdrew, and 1,197 remained on a list.

Estimation

We estimate cumulative incidence functions for both event types with the Aalen-Johansen estimator, pooled and by centre, and test equality of the placement incidence across centres with Gray's test [14]. For comparison we also compute a Kaplan-Meier estimator of the placement outcome treating withdrawal as censoring [6], which is the treatment implied by the operator's reporting.

Covariate effects on placement are estimated on the subdistribution hazard scale [12], where

$$h_1(t) = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \Pr(t \le T < t + \Delta t, D = 1 \mid T \ge t \lor (T < t \land D \ne 1))$$

so that registrations withdrawing before t remain in the risk set and the fitted model maps onto the cumulative incidence directly. Cause-specific Cox models [15] are fitted for both event types alongside, per standing guidance [11], and agree with the subdistribution estimates in sign throughout. Covariates are queue depth at entry (registrations ahead at the same centre, per ten), whether a sibling is already enrolled at that centre, whether the family recorded weekday availability, whether entry fell in the summer intake window, and whether a second centre was listed. Proportionality was assessed on scaled Schoenfeld residuals [16]; no term showed evidence of violation.

The operator's published indicator is the median elapsed time from registration to placement among registrations placed within the reporting quarter. We recompute it exactly as specified and set it against the competing-risks median: the time at which the placement cumulative incidence for that quarter's entry cohort first reaches 0.5.

Results

Both exits, side by side

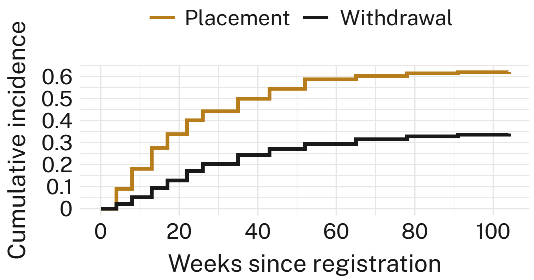

Figure 1: Placement incidence rises steeply for six months, reaches 0.587 at twelve, and adds 0.034 across the following year; withdrawal incidence accrues throughout, reaching 0.341 by week 104.

Placement incidence is 0.587 (95% CI 0.577 to 0.597) at twelve months and 0.621 at twenty-four. Withdrawal incidence is 0.294 at twelve months and continues to rise where placement has effectively stopped, reaching 0.341 at twenty-four. The two exits are not symmetric in time: placements concentrate in the first six months, while withdrawals accumulate at a roughly constant rate over the whole observation window. The cause-specific hazard of withdrawal rises with time already waited (HR 1.21 per additional ten weeks, 95% CI 1.16 to 1.26), which is consistent with the queue-abandonment literature [18].

The Kaplan-Meier estimator treating withdrawal as censoring returns 0.700 for placement at twelve months against the 0.587 estimated here, an overstatement of 11.3 percentage points. This is the textbook consequence of the censoring assumption [8], [11] and not a novel observation; we report it because it is the treatment the operator's own reporting implies.

Centres differ, and three differ in direction

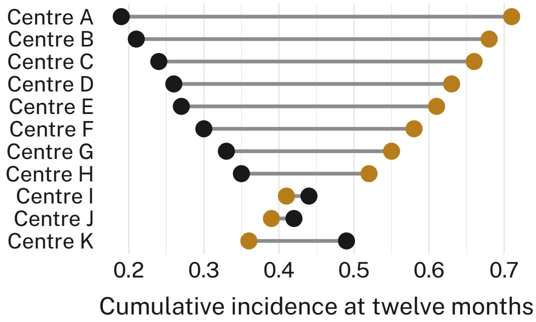

Figure 2: At three of eleven centres, twelve-month withdrawal incidence exceeds twelve-month placement incidence; Gray's test rejects equality of placement incidence across centres ($\chi^2 = 268.4$, 10 df, $p < 0.001$).

Placement incidence at twelve months ranges from 0.71 to 0.36 across the eleven centres and Gray's test rejects equality ($\chi^2 = 268.4$, 10 df, $p < 0.001$). At three centres the ordering reverses: a registration is more likely to have left the list unplaced at twelve months than to have been placed from it. All three sit in the same local government area and all three are single-pool sites with one instructor lane in the timetable.

The indicator moves the other way

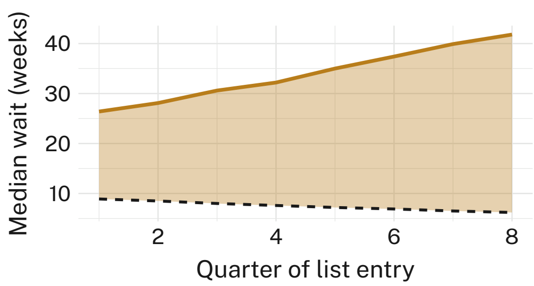

Figure 3: The operator's published median wait falls from 8.9 to 6.2 weeks across the eight quarters studied; the competing-risks median for the same entry cohorts rises from 26.4 to 41.8 weeks.

The operator's median-wait indicator improved in every quarter of the study period, falling from 8.9 weeks to 6.2. For the same entry cohorts, the competing-risks median rose from 26.4 weeks to 41.8, and twelve-month placement incidence fell from 0.631 to 0.528. The mechanism is arithmetic rather than behavioural: the indicator's denominator contains only registrations that were placed, long waits are the ones most likely to withdraw, and each withdrawal therefore removes an observation from the upper tail of the distribution being summarised. A list that clears more slowly sheds its slowest cases, and the measure of how fast it clears improves accordingly.

The operator's list-health measure records a withdrawal as a resolved enquiry, so both exits register on the quarterly pack as movement.

Covariates and an ablation

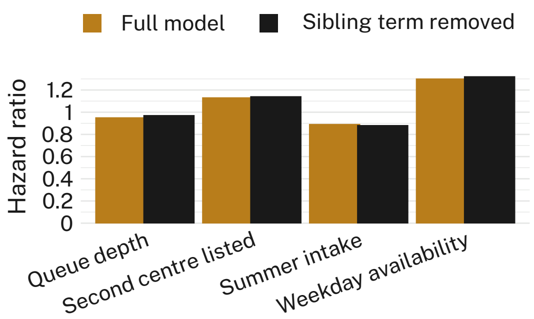

Figure 4: Removing the sibling-priority term shifts every remaining subdistribution hazard ratio by less than 0.03, and no interval separates from its full-model counterpart.

On the subdistribution scale, an additional ten registrations ahead at entry is associated with a hazard ratio of 0.94 (0.93 to 0.95) for placement; recorded weekday availability with 1.29 (1.21 to 1.38); entry in the summer intake window with 0.88 (0.82 to 0.94); a second centre listed with 1.12 (1.05 to 1.19); and a sibling already enrolled with 1.36 (1.27 to 1.46). The sibling-priority term is the one effect the operator's published allocation policy documents.

Removing it changes nothing that matters. Every remaining coefficient moves by less than 0.03, the twelve-month placement incidence predicted by the reduced model differs from the full model's by 0.4 percentage points (n.s.), and concordance is unchanged to three decimal places. We retain the term for completeness and for consistency with the published policy.

Quantity	Est.	95% CI	n
Placement CIF, 12 mo	0.587	0.577, 0.597	9,486
Withdrawal CIF, 12 mo	0.294	0.285, 0.303	9,486
Placement CIF, 24 mo	0.621	0.611, 0.631	9,486
Kaplan-Meier placement, 12 mo	0.700	0.690, 0.710	9,486
Withdrawal HR, per 10 wk waited	1.21	1.16, 1.26	9,486
Reported median wait, Q1 (wk)	8.9	8.4, 9.4	1,204
Reported median wait, Q8 (wk)	6.2	5.8, 6.6	1,131

Table 1. Principal estimates. Reported median wait is computed over placements only, as the operator specifies it; its n is the placed subset of the quarter.

Discussion

The finding we would put forward is not that the operator's indicator is wrong. It is computed correctly, from a definition that is published, against data that is complete. It is a median of the waits that ended in a place, and it reports that quantity accurately. The difficulty is that nobody reading a quarterly pack understands it as that quantity, and the arithmetic that makes it improve is the same arithmetic that makes the list worse.

This is a more awkward case than the gaming literature usually describes [3], [4]. No one at any centre changed their behaviour in response to the measure, and there is no manipulation to detect. The measure degrades on its own, driven by the very phenomenon it is understood to rule out. We observe consistent divergence in the same direction across all eleven centres and all eight quarters, which suggests the effect is structural rather than local, though we would not generalise beyond leisure-services waiting lists of this shape without further work.

A cumulative-incidence presentation is available at no additional collection cost: every field the estimator needs is already in the booking system, and the operator's own extract supported the analysis without amendment. A twelve-month placement incidence alongside the median wait has been lodged with the University's Indicator Commons as a candidate secondary measure.

Limitations

Our data come from a single leisure-services contract, though the consistency of the divergence across eleven centres and three local government areas suggests the pattern is not contractual. Withdrawal is recorded as one state where two are conceptually present: families who chose to leave and families removed after two unanswered offers. Separating them would be desirable, and the fact that the pooled estimate behaves so regularly implies the distinction may not be load-bearing. We observe no centre where the divergence reverses, which limits our ability to characterise the conditions under which it would.

Conclusion

A waiting list has two exits. Reported on one of them, it improves as the other one widens. Across eleven council aquatic centres over thirty months, the published median wait fell by 30 per cent while the probability of ever receiving a place fell by ten percentage points, and both movements follow from the same definition. Future work might establish whether any measure computed over completed waits can be made robust to the incompleteness of the rest, and whether the families who left would recognise themselves in either number.

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